

Math 242, S15
Quiz 10

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find Laplace transforms of the following functions:

(a) $f(t) = -2e^{3t}$

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \mathcal{L}\{-2e^{3t}\} = -2 \frac{1}{s-3} \\ &= -\frac{2}{s-3}, \quad s > 3.\end{aligned}$$

(b) $f(t) = \sin 3t + 2 \cos 3t$

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin 3t\} + 2\mathcal{L}\{\cos 3t\} \\ &= \frac{3}{s^2+3^2} + 2 \cdot \frac{s}{s^2+3^2} \\ &= \frac{2s+3}{s^2+9}\end{aligned}$$

Problem 2. (5 points) Find inverse Laplace transforms of the following functions:

(a) $F(s) = \frac{6}{s^4}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{6}{s^4}\right\} = t^3$$

(b) $F(s) = \frac{4}{s^2-4}$

$$\begin{aligned}\mathcal{L}^{-1}\{F(s)\} &= \mathcal{L}^{-1}\left\{\frac{4}{s^2-4}\right\} = \mathcal{L}^{-1}\left\{\frac{2 \times 2}{s^2-2^2}\right\} \\ &= 2 \sin 2t\end{aligned}$$